Global to Local: Clip-LSTM-Based Object Detection From Remote Sensing Images

Zhu Teng, Yani Duan, Yan Liu, Baopeng Zhang, and Jianping Fan

***Abstract*— Object detection from remote sensing images (RSIs) is a basic topic in the area of aviation and satellite image processing, which has great effects on geological disaster detection, agricultural planning, and land utilization. However, it is always faced with several severe difficulties. For instance, the scale of the target spans over a very wide range, and the difference between the target size and the image size is huge, as some targets only account for a dozen pixels compared with the RSI of the megapixel level. In this work, an innovative object detection network (GLNet) is proposed for remote sensing imagery. Our approach integrates global context clues extracted by the multiscale perception (MSP) module and local spatial contextual correlations encoded by Clip long short-term memory (Clip-LSTM). These rich semantic features are further exploited to design a self-adapted anchor (SA) module to alleviate the scale variations in RSIs. Extensive experiments are executed on several public easily accessed benchmarks, including DOTA, NWPU VHR-10, and DIOR. Experimental results have demonstrated that our GLNet outperforms numerous latest methods.**

***Index Terms*— Clip long short-term memory (Clip-LSTM), global context, local spatial correlation, object detection, remote sensing imagery.**

I. Introduction

With the continuous development in technologies of remote sensing images (RSIs) and image spatial resolution, the amount of data for RSIs has increased dramatically. As a significant issue in aerial image processing and analysis, object detection from RSIs is extensively employed in many applications, such as environmental detection, geological disaster detection, precision agriculture, land use, and so on [1]–[5]. Massive image data not only provide a significant data source for object detection from RSIs but also bring huge challenges.

Traditional object detection approaches mainly design handcrafted features based on human experiences [6]. However, these human intervened methods heavily rely on experiences and prior knowledge, which results in limited adaptability and generalization ability of the detection methods. In recent years, deep convolutional neural networks (CNNs) are extensively used in the object detection field because they can actively learn features from data rather than designing them according to artificial experiences.

Compared with the ordinary detection task, object detection from RSIs has its specific attributes. The first one is the tremendous difference between the target size and the image size in RSIs. In an RSI with megapixels, some targets only occupy a dozen pixels. To this end, we design a multiscale perception (MSP) module to exploit the latent global contextual information in multiple layers with different resolutions in the feature extraction phase. On the one hand, it alleviates the lack of texture details in RSIs to some extent. On the other hand, the encoding of the overall scene (global features) may facilitate the understanding and recognition of specific targets since targets in RSIs may correlate with specific scenes. For instance, in a coast scene, ships may call in at where port exists.

Second, the characteristics of targets in RSIs are mainly reflected in contours. In contrast to general object detection tasks, the targets in RSIs lack rich texture information, which complicates the entire inspection task. In fact, the local contexts in RSI are conducive to the recognition of the targets, especially in the situation of densely allocated targets. For example, the presence of other vessels close to a predicted bounding box has a significant indication of the existence of the vessel at the current bounding box. Therefore, we design clockwise connected long short-term memory (LSTM) units to capture the spatial information of the target and encode the local connections between multiple LSTM units in different resolutions. The exploitation of the global and local contextual information provides strong representation ability for the detection task in RSIs.

Finally, the sizes of targets for the same class are often distributed over a very wide range. Take ships as an example. It is very challenging to detect various types of ships with different scales. With regard to this, a self-adaptation anchor (SA) module is developed to generate unconstrained anchors based on image semantic features. With the expressive global and local features, the anchor generation stage is more precise and efficient, laying a solid foundation for subsequent stages. Two branches construct SA to predict the possible locations and sizes of anchors. For the sake of efficient calculation, we adopt a bottleneck structure in SA to amplify the receptive field of the feature maps relative to the original image. The SA module not only facilitates the multiscale object detection in RSIs but

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also enhances the adaptability and generalization capacity of our method.

To sum up, we propose an object detection network named GLNet for RSIs, which consists of three modules, including an MSP module, a Clip-LSTM module, and an SA module. MSP encodes the image from a global perspective and extracts the global contextual information with different resolutions. The Clip-LSTM module exploits the spatial correlation information between local parts of targets at multiple appropriate scales. The SA module is developed based on image semantic features. It employs a bottleneck structure to amplify the receptive field, which enhances the capabilities of the small target detection. The overall GLNet integrates and learns both global contextual information and local correlation information in an end-to-end form.

We make the following arrangements for the remainder of this work. In Section II, the related literature on object detection is summarized. The proposed GLNet is described in Section III. In Section IV, we detail our implementation, report comparison results on several widely employed benchmarks, and conduct ablation studies on the components of GLNet. We come to conclusions in Section V.

II. Related Work

As a fundamental and critical topic in remote sensing, object detection technology for RSIs has been continuously improved in the last few years. First, we briefly investigate object detection in natural images and then discuss detection methods for RSIs.

A. Object Detection in Natural Images

Deep learning technology has achieved a series of remarkable results in the object detection area. Employing deep learning technology, methods for target detection in natural images are mainly classified into two categories, methods based on region proposals and the regression-based object detection approaches.

In the proposal-based methods, the region proposal generator plays a key role, and whether it can produce a high-quality region of interest (ROI) influences the detection accuracy remarkably. For instance, RCNN [7] generates about 2000 regional proposals by a selective search algorithm and constructs CNNs to extract features on each regional proposal. Fast-RCNN [8] enhances RCNN in the details of feature extraction and detection and utilizes the softmax layer instead of support vector machine (SVM) for prediction, which requires only one pass on the entire image. Based on this model, Faster-RCNN [9] is proposed by adopting an RPN network to produce regional proposals, rather than employing the slower selective search approach. R-FCN [10] puts forward a position-sensitive score map to predict the class and location of the entire object.

In the regression-based methods, the proposal generation is not required and bounding box regression is directly operated. There are two representative methods, including SSD [11] and YOLO [12]. SSD employs a fully CNN to detect objects on feature maps of several convolutional layers and generates anchors with several predefined sizes and length-to-width ratios to predict the category score. YOLO does not assume the objects that come from predefined anchors but divides the image into several grids to predict bounding box positions and corresponding confidence scores. Besides YOLO, anchor-free approaches have been greatly promoted recently. For instance, CornerNet [13] represents objects by the top-left and bottom-right corners of the object rather than anchors. CenterNet [14] represents each object by an additional center key point and a pair of corners, which improves both precision and recall. ExtremeNet [15] employs a standard key-point estimation network to detect four extreme points and a center point of the object. While FCOS [16] and FoveaBox [17] regress a vector of object geometry (such as boundaries of the object or its distance to the borders) to determine the shapes of objects. These above methods obtain a remarkable accuracy in object detection. However, straightly applying generic target detection methods to the field of RSIs deteriorates detection performance due to the widely distributed length-to-width ratios of the targets and the tremendous difference between the target size and the image size in RSIs.

B. Object Detection in RSIs

During the past few decades, considerable efforts have been devoted to object detection in RSIs. Traditional object detection methods from RSIs can be divided into four main categories: template matching-based methods, knowledge-based methods, methods based on object-level image analysis, and methods based on machine learning [18], [19]. Various templates are employed to detect specific objects in RSIs, such as road [20], buildings [21], and so on. Template matching-based methods are effective in some applications, but it requires a precise template, which is not practical in many scenarios, especially for viewpoint changes. Knowledge-based detection methods generally need to transfer the latent knowledge of objects into the explicit detection rules. For example, Ok [22] exploited the shadow evidence for building detection. However, the definition of rules is nontrivial and is not easy to apply extensively. Methods based on object-level image analysis usually first segment pixels into groups and then apply classification [23]. These methods require an accuracy measurement, and it is difficult to specify a general measure to evaluate the classification accuracy for all objects. Machine learning-based methods consider object detection as a classification problem, where features (such as histogram of oriented gradients (HOG) [24], texture [25], and so on) and classifiers (such as SVM [26], AdaBoost [25], artificial neural network (ANN) [27], and so on) are the most important two components. As deep neural networks achieve great success in object detection for natural images, many object detection methodologies for RSIs are directly migrated from object detection neural networks. Faster R-CNN architecture [28] is employed to detect targets in RSIs. Chen *et al.* [29] proposed a detection method with a variable input image scale on the SSD detection framework. Xia *et al.* [30] employed YOLOv2 [31] to detect objects in RSIs. Deng *et al.* [32] proposed a multiscale target detection architecture for RSIs. An airport

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detection algorithm is developed in [33] by using a cascading regional recommended network. As most of these methods are only aimed at the detection of single-class targets in RSIs, they do not sufficiently consider the global information and local connections of RSIs, which declines the overall detection performance.

Meanwhile, some innovative structures designed for RSIs are proposed recently. For example, in [34], a deconvolution layer is employed after the region proposal step, which reenlarges the size of targets. However, the effectiveness of deconvolution is restricted by small targets in RSIs due to the limited receptive field it receives. Dense feature pyramid networks are developed in [35], which uses RPN networks to extract feature maps of different layers. In this case, the selection of layers depends on some specific data set, which weakens the generalization ability of the model.

In particular, for targets with various scales in RSIs, traditional methods often integrate multiscale feature information of CNNs to enhance the performance. For instance, the spatial pyramid network structure is adopted in [36], which is a direct and effective way to dispose of the multiscale issue. In [37], the original feature representation of the target is utilized to decide which layer of ROI features should be used. Features from multiple layers of the image pyramid are encoded in [38] to deal with this issue. A multiscale structure is employed [39] to adapt the scale variation, which contains a self-coding machine equipped with jump links, and it forecasts different scales by using different anchor frames. However, different sizes of anchor frames still need to be determined according to the inspected data set. Although these works address the multiscale issue in RSIs, they only distinguish the multiscale objects within limits and have little improvements on the regional proposal. It is not easy to adapt the variation of object sizes from various remote sensing benchmarks. Different from these works, our proposed method develops an MSP module to aggregate information in different resolutions and designs an SA scheme to acclimate to the variations of scales for the objects in RSIs.

On the issue of small targets ubiquitously existing in RSIs, numerous algorithms have been proposed. The algorithm YOLT [40] increases the number of grids in YOLO so that each grid can predict smaller targets. However, the ability of small target prediction is compromised because it is impossible to increase the number of grids infinitely. A coarse-to-fine detection method is presented in [41], but two CNN networks are required, which increases computational costs. In [42], the dilation convolution operation is utilized to expand the receptive field, but it brings about the loss of local target information. In this article, we employ a bottleneck structure to amplify the receptive field and encode the spatial information and local connections by the proposed Clip-LSTM module to facilitate the detection of small targets in RSIs.

Furthermore, RSIs also involve complex backgrounds and suffer from insufficient information on objects. In this respect, Zhang *et al.* [43] used the full convolution network to train and extract the semantic features on ocean, ship, and land, aiming to separate the sea from land exactly. In [44], a large number of negative samples that have no targets are trained to suppress the influence of targets with similar characteristics to ships under the land background. In [45], a position-sensitive score graph is employed after the region proposal stage to localize the target. Different from these approaches, our model integrates the global and local context information to resist the background clutters and exploits the Clip-LSTM module to handle the insufficient information issue.

III. GLNet

We collect clues from global to local and propose a robust and accurate approach to detect targets from RSIs. The architecture of our GLNet is presented in Fig. 1 where three modules are involved. The MSP module brings in the attention mechanism to leverage different areas so that the module pays more attention to regions with rich and discriminative information. The Clip long short-term memory (Clip-LSTM) module consists of several LSTM units, and each LSTM unit encodes different spatial information. The entire Clip-LSTM exploits local connections between different target regions, which incorporates local contexts. The SA module designs unconstrained anchors under the assistance of image semantic features, and proposals are parameterized relative to these anchors. These three modules can be used independently and can be perfectly combined with the existing object detection algorithms for RSIs.

A. MSP Module

The attention mechanism has become a significant part of CNNs and is used in many applications across multiple fields. It can be explained directly by the human visual system, which often ignores irrelevant information and focuses on the information we are interested in. From this point of view, many works adopt “mask" to combine the visual attention mechanism with deep neural networks, and it weighs the importance of features and makes the network devoted to the more significant information rather than the irrelevant information.

Our proposed MSP module constructs a multilevel attention mechanism to identify and extract more useful global contextual information from RSIs in the feature extraction stage. The structure of the MSP module is presented in Fig. 2 where the attention mechanism is exploited on several levels of feature maps in different resolutions. It captures more comprehensive features and understands the image from the perspective of “global” rather than being caught into the dilemma of seeing some local spots. It is laborious for a traditional convolution layer to implement “global” by itself because it mines the pixel relationships among local neighborhoods. In contrast, our proposed MSP module aggregates global information from several layers with different resolutions. The design of the MSP compensates for the information loss of small targets in traditional CNN architectures and suppresses background noise.

Assume that the attentional multiresolution feature maps (as described in Fig. 2) are represented by $F_0 \sim F_3$, and the output multilevel feature maps are denoted by $M_0 \sim M_3$ from top to bottom. The relationship between two consecutive levels of

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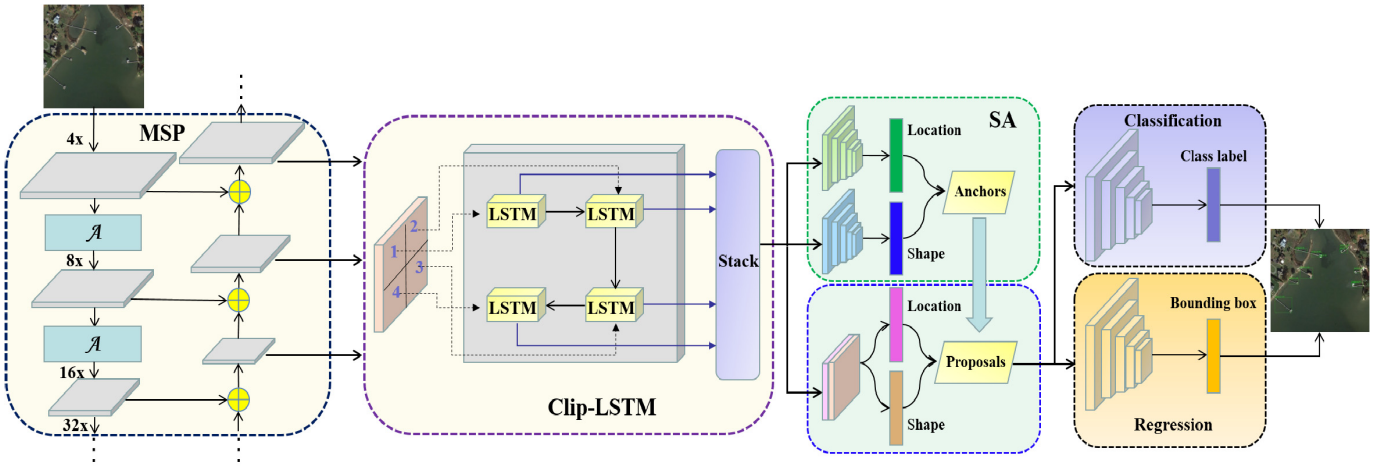


Fig. 1. Architecture of our proposed GLNet. Feature maps with MSPs in multiple layers are forwarded to the Clip-LSTM module to integrate the global context information and local spatial correlations. Based on this, the SA module predicts the shape and location score of anchors, and proposals are parameterized to these anchors. Finally, the class label and bounding box are determined by the corresponding prediction module.

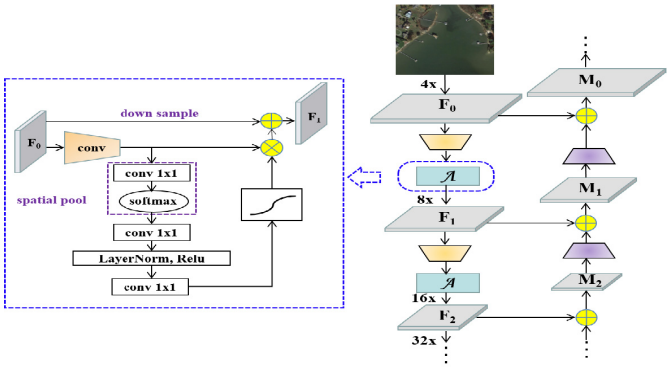


Fig. 2. Structure of the MSP module. The orange ladder denotes the downsampling operation, and the purple ladder represents the upsampling operation. The box on the left shows in detail how the attention mechanism is constructed in our work.

attentional feature maps is described by (1), where Ψ refers to the downsampling operation and $\mathscr{A}$ represents the attention operation, and four levels of feature maps are employed in our module, i.e., $i \in \{0, 1, 2\}$

$$F_{i+1} = \mathscr{A}(\Psi(F_i)) \tag{1}$$

$$\mathscr{A}(F_i) = \sigma(\phi(F_i)) \odot \Phi(F_i). \tag{2}$$

The attention operation is expanded in (2), where σ represents the sigmoid function, ϕ denotes for spatial pooling operation, the convolution operation is expressed by Φ, and $\odot$ stands for the elementwise multiplication as presented in Fig. 2. The conv 1×1 operation in spatial pooling operation is used as a dimension reduction and transformation on feature maps.

The output of the MSP module is further enhanced by the attentional feature maps as expressed by (3) with $i \in \{2, 1, 0\}$, where ψ represents the upsampling operation. Note that the ground level of input feature maps is transferred to the output feature map, i.e., $M_3 = F_3$. The output feature maps $M_0 \sim M_3$ are finally upsampled to the same size and aggregated by concatenation, after which we obtain the global context information for the Clip-LSTM module

$$M_i = F_i + \psi(M_{i+1}). \tag{3}$$

To testify the effectiveness of the MSP module, feature maps processed by this module are visualized on several RSIs, as shown in Fig. 3. The brighter the color is, the more attention it obtains. It can be observed from Fig. 3 that our proposed MSP module continuously focuses on the target region at different levels of feature maps and suppresses the background interference.

B. Clip-LSTM

The LSTM structure is beneficial in language translation, computer vision, and other tasks because it can eliminate or enhance information in feature representation by a designed memory mechanism. This guarantees that the sequential information can be captured from the input data. Inspired by this, we take full advantage of this memory mechanism to remember the relationship among local regions in remote sensing imagery. Furthermore, the LSTM network can extract the spatial contextual information of the target.

We design a Clip-LSTM module based on the observation that the region around the target provides some useful context information to infer the target itself. For example, the possible targets (candidates) are more likely to be ships rather than airplanes if the surrounding context is the sea. The detection and recognition of new targets are enhanced by encoding the interaction information between different parts of the image and the interdependent correlation between objects and scenes. The context information exploited by our model is not directly gained from the appearance of the target, but the neighborhood information surrounding the target.

Our proposed Clip-LSTM module first cuts crosses on the ith scale level of feature maps ($M_0 \sim M_3$, $i \in \{0, 1, 2, 3\}$), and several local patches (S represents the total number of local patches) are generated, which are denoted by $P_1^i \sim P_S^i$,

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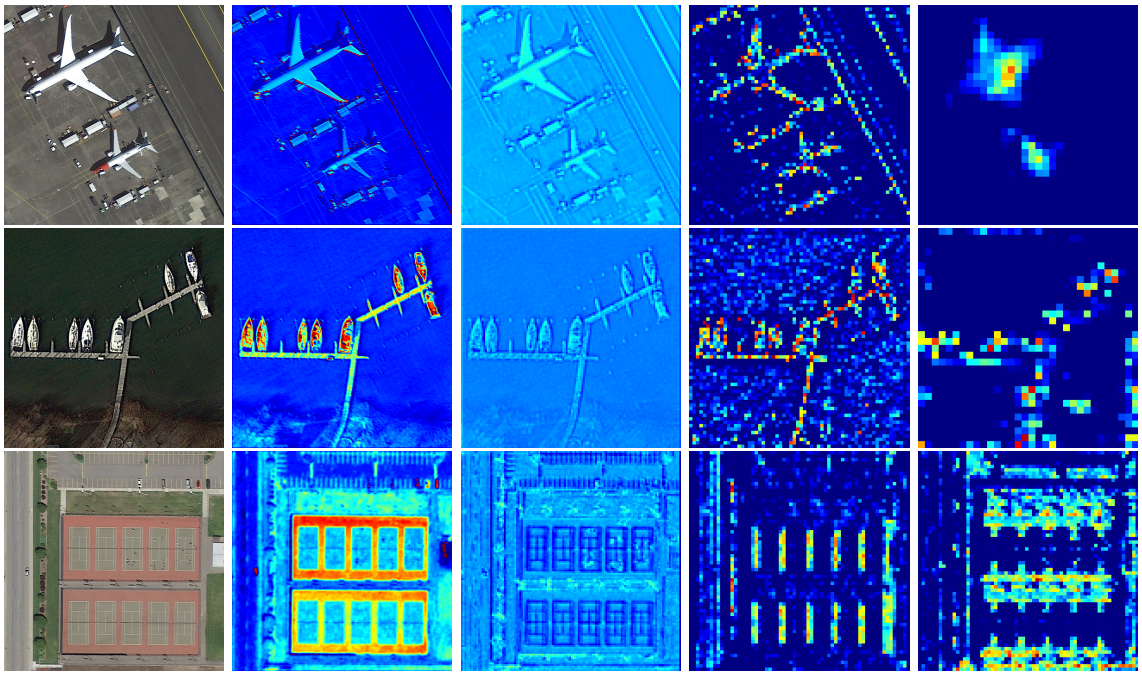


Fig. 3. Visualization of feature maps processed by the MSP module on several RSIs. (a) Original Image. (b) Layer I. (c) Layer II. (d) Layer III. (e) Layer IV.

respectively. Then, each local patch is conveyed into an LSTM unit. In a clockwise manner, the LSTM unit propagates this local patch and the hidden state of its previous spatial patch and applies an update function to refresh the hidden state. The overall Clip-LSTM aims at capturing the interaction information between these local patches. We denote that the propagation function as Ω and the LSTM unit of our Clip-LSTM module can be formulated by (4) where $s = 1, 2, \ldots, S$

$$h_{s+1} = \Omega\left(h_s,\ P^i_{s+1}\right). \tag{4}$$

The output of each LSTM unit is finally stacked together to form a more expressive feature map, and Clip-LSTM searches for the spatial contextual relationship among local patches. The newly formed feature map is further sent to the SA module (see more details in Section III-C) for subsequent prediction. Besides, other RNN architectures such as GRU can also be used in our GLNet. GRU replaces the forget gate and input gate with the update gate and combines the cell state and the hidden state to reduce memory and raise efficiency. It has fewer parameters and requires less training data, while in the situation of sufficient training data, the LSTM architecture has better representation ability. There are also some other architectures such as convolutional LSTM (ConvLSTM) that combines sequential and spatial information. Although our Clip-LSTM module and ConvLSTM can encode the spatial information of the image, they are different. The ConvLSTM determines the future state of a certain cell in the grid by the inputs and past states of its local neighbors, which can be easily achieved by using a convolution operator in the state-to-state and input-to-state transitions. In contrast, the Clip-LSTM consists of several clockwise connected LSTM units, and the spatial information is captured by both the local neighbors encoding via a single LSTM unit and the local connections between multiple LSTM units. Through this designed LSTM module, the contextual patches are highlighted iteratively, which seeks the correlation between different local regions and provides a strong feature representation for object detection. The proposed structure is extensible, depending on the number and type of LSTM units, and an ablation study on the number of LSTM units is conducted in Section IV-C.

C. SA Module

It is widely known that the length-to-width ratios of targets in RSIs vary greatly and numerous data sets exist in the field of detection from RSIs. To this end, we need a robust and accurate detector to localize multiscale targets from diverse data sets of RSIs. In general, most methods artificially preset several anchors with fixed sizes, and the final prediction results are produced by forwarding the network at anchor frames. Since the anchors are often randomly generated or artificially presetted, it may be inappropriate for the targets with large-scale changes. Furthermore, it also leads to low efficiency, a large proportion of negative anchor frames, and difficulties in adapting to the scale variations across multiple data sets.

This article develops a self-adaptive anchor generation method, named SA, which can generate precisely adaptive anchors based on image semantic information. The architecture of the SA module is presented in Fig. 4 where two branches, including the location prediction and the shape prediction, are employed to predict the center and the width as well as the height of unconstrained anchors, respectively. The

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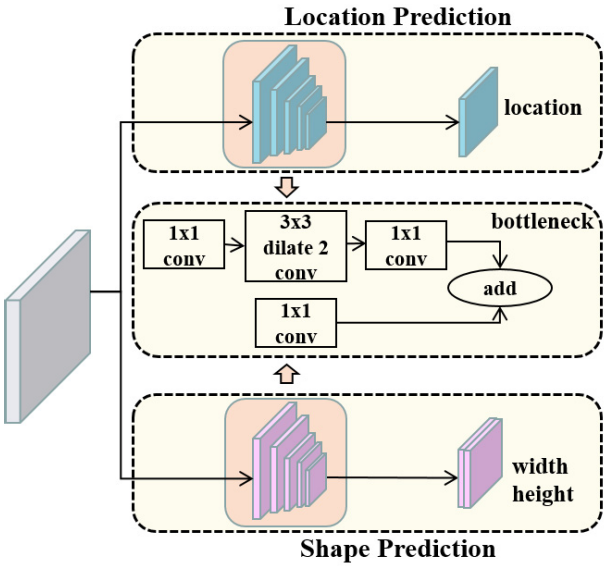


Fig. 4. Architecture of the SA module. The center and the shape of the objects are, respectively, estimated by the branch of location prediction and shape prediction.

shape and location predictions are executed on features with different scales separately. To facilitate the discovery of small targets in RSIs, we further adopt the bottleneck structure to amplify the receptive field of feature maps in the prediction of both location and shape of unconstrained anchors. The spatial resolution is not changed by the bottleneck structure, but the number of channels is different for the location and shape branch. Finally, the unconstrained anchors are produced by combining the corresponding location and shape. The design of SA not only alleviates the target scale variation problem in RSIs but also enhances the generalization ability as well as the adaptability of multiscale targets across a variety of data sets.

Next, we elaborate on the network architecture of the SA module. In general, SA consists of two bottleneck branches, named the location branch and shape branch. The location branch is to estimate the possible center locations of anchors in RSIs, and the shape branch forecasts the possible width and height of the anchors. There are three 1×1 convolution layers and one 3×3 dilated convolution in each bottleneck. The dilated convolution can feed the subsequent convolution layer with larger feature maps by not expanding the number of parameters, which is beneficial to the recognition of small targets. The two 1×1 convolutions before and after the dilated convolution are used to reduce and increase the dimension to obtain the prediction score of the location and the size. The below extra 1×1 convolution is mainly used to take full advantage of residual features in the original image to enhance the detection accuracy.

D. End-to-End Network Learning

Until now, we described our proposed method from the perspective of architecture. In this section, our loss function and end-to-end learning are discussed. In the meticulously designed feature extraction network, ResNet-101 is endorsed as our backbone, and the attention mechanism is added into several layers of ResNet-101 to enable the feature network to concentrate on the regions with rich information. Our MSP module congregates precise location information in shallow layers and rich semantic features in deep layers, which is profitable for the detection of various objects in RSIs. The parameters of the MSP module are reported in Table I where conv3, conv4, and conv5 are appended with the attention mechanism.

TABLE I
PARAMETERS OF THE MSP MODULE

Layer name	Building blocks	Number of blocks
conv1	7×7, 64	1
conv2	1×1, 128 3×3, 128 1×1, 256	3
conv3	1×1, 256 3×3, 256 1×1, 512	4
conv3_attention	1×1, 1 1×1, 128 1×1, 512	
conv4	1×1, 512 3×3, 512 1×1, 1024	23
conv4_attention	1×1, 1 1×1, 256 1×1, 1024	
conv5	1×1, 1024 3×3, 1024 1×1, 2048	3
conv5_attention	1×1, 1 1×1, 512 1×1, 2048	

In the design of the loss layer in the GLNet, a multitask joint loss function containing six terms is proposed and we construct an end-to-end learning strategy to optimize GLNet. The six terms are originated from loss on the location prediction and shape prediction in the SA module, proposal classification, proposal regression, object classification, and object regression as described in (5). We fully consider the functionality of each term in the joint loss function at the same time

$$L_{\text{joint}} = \alpha \sum_{\upsilon}^{\Upsilon} L_{\text{SA_pos}}^{(\upsilon)} + \beta \sum_{\upsilon}^{\Upsilon} L_{\text{SA_size}}^{(\upsilon)} + \sum_{\upsilon}^{\Upsilon} L_{\text{Pro_cls}}^{(\upsilon)} + \sum_{\upsilon}^{\Upsilon} L_{\text{Pro_reg}}^{(\upsilon)} + \sum_{\upsilon}^{\Upsilon} L_{\text{Obj_cls}}^{(\upsilon)} + \sum_{\upsilon}^{\Upsilon} L_{\text{Obj_reg}}^{(\upsilon)} \quad (5)$$

where Υ represents the total number of samples in a minibatch and υ denotes the sample index currently being processed. The first term $L_{\text{SA_pos}}^{(\upsilon)}$ restricts the location prediction and a focal loss is utilized. The second term $L_{\text{SA_size}}^{(\upsilon)}$ employs the smooth L_1 loss on the shape branch in the SA module. The third term $L_{\text{Pro_cls}}^{(\upsilon)}$ denotes for the classification loss constrained on proposals, and the fourth term $L_{\text{Pro_reg}}^{(\upsilon)}$ represents the regression loss on the proposal generation. The last two terms are classification and regression loss imposed on the object prediction level.

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TABLE II
SPECIFIC CATEGORIES OF DOTA, NWPU VHR-10, AND DIOR

Datasets	Classes				
DOTA	plane (P)	ground track field (GTF)	storage tank (ST)	baseball diamond (BD)	tennis court (TC)
	ship (S)	soccer ball field (SBF)	small vehicle (SV)	basketball court (BC)	large vehicle (LV)
	harbor (H)	swimming pool (SP)	roundabout (RA)	helicopter (HC)	bridge (B)
NWPU VHR-10	airplane	baseball diamond	basketball court	bridge	harbor
	ship(S)	ground track field	storage tank	tennis court	vehicle
DIOR	airplane	expressway service area	baseball field	basketball court	bridge
	airport	expressway toll station	golf course	storage tank	harbor
	chimney	ground track field	tennis court	train station	overpass
	dam	stadium	windmill	vehicle	ship

In particular, on the anchor level, the focal loss and smooth L_1 loss function aim to decrease the unrepresentative data in the training process and balance the proportional value of negative and positive samples. On the proposal and object level, the classification loss employs cross entropy, whereas the regression loss utilizes the smooth L_1. These two loss functions define the approximation degree between the expected output value and the output value of the actual network.

IV. EXPERIMENTAL RESULTS

The details and results of our experiments are elaborately discussed in this section, including implementation details in Section IV-A, experimental results performed by our GLNet and numerous state-of-the-art approaches in Section IV-B, and ablation study in Section IV-C.

A. Implementation Details

a) Data Sets: We adopt three data sets of RSIs to verify the effectiveness of GLNet, the following is a detailed description of them, and the specific categories of each data set are described in Table II.

1) DOTA [30] is a large-scale RSI data set for aviation images. The DOTA-v1.0 contains 2806 aerial images from simultaneous interpreting of distinct sensors and systems and includes 188 282 targets marked by an arbitrary quadrilateral with eight degrees of freedom. The range of image size spans from 800 × 800 or so to about 4000 × 4000 pixels, and it includes various scales, directions, and shapes of targets. The DOTA pictures are labeled by professionals.
2) NWPU VHR-10 [46] is a ten-level data set for geographical RSIs used for space object detection. It contains 650 positive images and 150 negative pictures without targets, a total of 800. The samples are derived from Google Earth and Vaihingen data sets. After cropped, they are manually labeled by professionals.
3) DIOR [19] is a publicly accessible massive optical RSI data set for object detection. It is the largest data set considering the magnitude of pictures among the current data sets. It consists of 23 463 images collecting from 20 object categories. The size of images in DIOR varies widely, not only in terms of the spatial resolution of the entire images but also in terms of intercategory and intracategory objects.

b) Experiment Setting: All the experiments in this work are realized via Pytorch deep learning framework in a Ubuntu operation system and conducted on the server with NVIDIA Geforce GTX 1080 Ti. The proposed method employs a synchronized stochastic gradient descent to optimize the training and test phase and support both oriented bounding boxes (OBBs) and horizontal bounding boxes (HBBs) as ground truth. The initial learning rate, momentum, and weight decay are, respectively, set to 0.0025, 0.9, and 0.0001. In the training phase, we executed 12 epochs in total, and starting from the eighth epoch, the step learning rate is updated. At the proposal generation stage, final classification stage, and regression stage, we all set the nonmaximum suppression (NMS) threshold to 0.7. The ratio of a positive proposal is set to 0.5 in the proposal generation stage. The main purpose is to obtain more precise proposals from the anchors. In the final prediction stage, the ratio of a positive bounding box is set to 0.25 to produce more accurate target results.

B. Experimental Results

We compare our GLNet with several state-of-the-art detectors on three widely used benchmarks, including DOTA, NWPU VHR-10, and DIOR. The performance on each category is examined by average precision (AP). The overall performance on a data set is estimated by the mean value of average precision (mAP) on all categories of the corresponded data set.

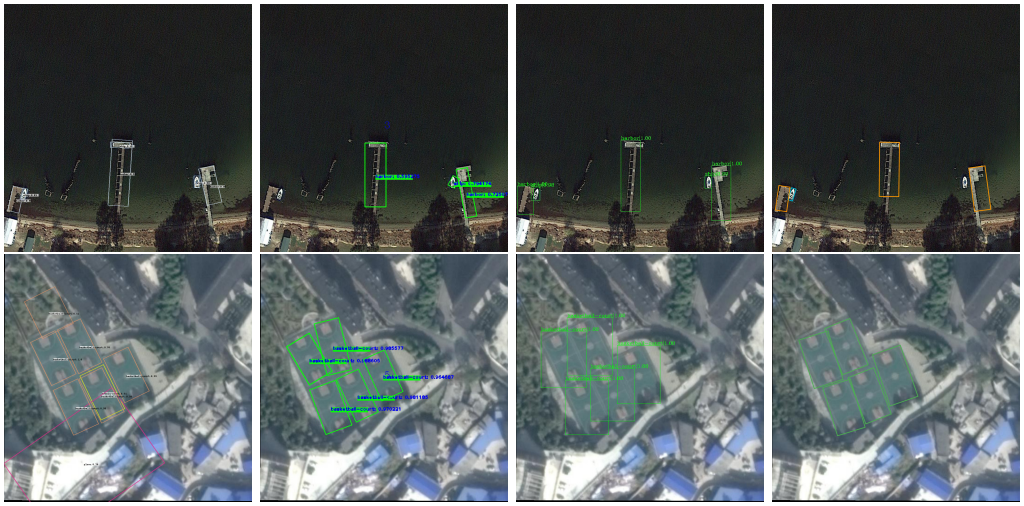
Results on DOTA: We evaluate the GLNet on DOTA with comparisons to numerous latest detectors, including IENet [47], R-DFPN [48], RRPN [49], CAD-Net [50], R3Det [51], RICNN [52], USB-BBR [53], Coarse-to-Fine [41], Faster RCNN [9], Yolo v3 [54], and MS-VANs [39]. Among these methods, IENet, R-DFPN, RRPN, CADNet, and R3Det employ OBBs as ground truth, and the others are based on HBBs. The proposed GLNet can use both HBB and OBB as ground-truth. The OBB version of GLNet is implemented by only augmenting an output of the oriented angle of the object in the regression module, i.e., an oriented rectangle. Note that HBB annotates only two points for each target, and OBB requires a four-point annotation for each target, which doubles the annotation efforts. All of these detectors

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TABLE III

QUANTITATIVE COMPARISONS BETWEEN OUR GLNET AGAINST STATE OF THE ARTS ON DOTA. THE PERFORMANCE IS MEASURED BY AP (%)

	Methods	P	S	ST	BD	TC	BC	GTF	H	B	SV	LV	SBF	RA	SP	HC	mAP
OBB	IENet [47]	80.2	52.6	78.5	64.5	81.5	44.7	32.1	64.4	39.8	49.7	65.0	46.5	56.7	64.2	36.8	57.1
	R-DFPN [48]	80.9	54.8	68.7	65.8	90.3	66.3	58.9	55.1	65.8	55.8	50.9	48.7	51.8	51.3	35.9	57.9
	RRPN [49]	88.5	57.3	67.4	71.2	90.8	72.8	59.3	53.1	71.2	51.9	56.2	56.7	52.8	51.9	53.6	61.0
	CAD-Net [50]	88.3	76.7	74.2	71.7	90.8	77.3	66.5	65.7	51.4	72.4	64.5	45.9	60.2	56.7	48.3	67.4
	R3Det [51]	89.5	77.5	83.5	82.0	90.8	81.4	62.5	65.4	48.5	70.5	74.3	62.0	59.8	67.5	60.1	**71.7**
	Ours-OBB	88.3	76.5	85.2	79.5	89.4	83.5	64.7	64.5	44.2	71.0	70.8	53.4	66.0	67.0	57.1	70.8
HBB	RICNN [52]	62.1	41.3	27.1	26.0	51.7	43.1	28.5	31.5	20.1	30.4	28.4	25.3	28.1	26.5	13.4	32.2
	USB-BBR [53]	69.2	46.3	29.3	28.6	55.7	45.1	30.3	34.1	19.5	33.5	29.6	27.4	31.9	33.5	14.2	35.2
	OFIC [41]	71.7	48.2	31.5	30.6	59.2	49.1	33.5	35.1	20.5	36.9	31.1	30.7	34.8	35.1	15.3	37.6
	Faster R-CNN [9]	78.4	47.2	58.6	76.8	89.7	74.3	67.3	60.8	32.1	51.9	51.5	55.3	48.7	55.2	40.8	59.2
	Yolo v3 [54]	79.0	49.8	59.2	77.1	89.9	74.8	68.1	61.5	33.9	52.8	52.2	55.5	49.0	55.9	41.7	60.0
	MS-VANs [39]	79.2	52.6	60.1	77.4	90.3	75.1	68.5	61.7	34.1	53.0	52.4	55.8	49.1	56.2	42.3	60.5
	Ours-HBB	89.4	53.9	77.2	71.5	79.9	66.9	68.4	73.1	43.1	31.9	52.5	74.1	64.6	59.2	74.0	**65.3**



(a) (b) (c) (d)

Fig. 5. Visualization of detection results by comparing GLNet against R3Det [51] and RRPN [49] on DOTA. (a) R3Det. (b) RRPN. (c) GLNet-HBB. (d) GLNet-OBB.

are CNN-based object detectors, among which MS-VANs is the latest HBB detector for RSIs that also employs multiscale feature maps. It provides a baseline for the multiscale feature involved detector. The results are reported in Table III. Our HBB model obtains an mAP of 65.3% averaged on all categories of DOTA, which outperforms the followed best HBB method MS-VANs by a remarkable gain (+4.8% mAP), indicating its high accuracy and robustness on the DOTA data set. In particular, it achieves the highest performance in the categories of small targets. For example, it obtains an AP of 89.4% for category plane (P), 53.9% for category ship (S), 43.1% for category bridge (B), and 73.1% for category helicopter (H), which suggests that our GLNet can detect small targets accurately even provided with limited information. The main reason is the design of the bottleneck structure and the Clip-LSTM. At the same time, concerning the large targets, such as harbor (H), storage tank (ST), roundabout (RA), and soccer ball field (SBF), our model achieves mAPs of 77.2%, 73.1%, 74.1%, and 64.6%, respectively. It ranks number one among all HBB methods and boosts the second-best HBB detector MS-VANs by a huge margin of 17.1%, 11.4%, 18.3%, and 15.5% on categories of ST, H, SBF, and RA, respectively. This superior performance on both small targets and large targets demonstrates that our model presents a good detection of multiscale targets in RSIs. Among all the OBB detectors, R3Det occupies the first place, and our GLNet-OBB ranks the second place, followed by CAD-Net. Experimental results also show that our OBB version outperforms our HBB version by a margin of 5.5%, and a possible reason is that OBB provides more effective ground-truth information than HBB because OBB eliminates most background information within the ground-truth bounding box. Some detection comparisons are visualized in Fig. 5. We can also notice from the table that our detection accuracy of several categories is lower than that of the detectors being compared, such as small vehicles. We analyze that it might be related to the distribution of targets. Our detectors are not robust enough for some orderly and densely arranged targets. On the contrary, for the strategies

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TABLE IV
QUANTITATIVE COMPARISONS BETWEEN OUR GLNET AND LATEST METHODOLOGIES ON NWPU VHR-10. THE PERFORMANCE IS EVALUATED BY AP (%)

	Methods	airplane	storage tank	baseball diamond	tennis court	basketball court	ground track field	harbor	bridge	vehicle	ship	mAP
OBB	CAD-Net [50]	97.0	95.6	93.6	87.6	87.1	99.6	100	86.2	89.9	77.9	**91.5**
HBB	MIF-CNN+ [55]	96.7	81.3	85.4	68.3	54.5	65.7	73.7	47.7	63.1	65.4	70.2
	Faster R-CNN [9]	90.9	77.9	90.9	90.3	81.8	88.6	80.3	70.4	78.8	80.2	83
	F-Faster R-CNN [56]	99.5	68.9	99.3	89.3	90.5	88.6	88.2	81.5	75.7	80.5	86.2
	DAPNet [56]	99.9	78.9	90.9	98.7	89.6	90.3	85.6	79.4	80.3	80.8	87.4
	SAPNet [57]	97.8	67.2	94.8	99.5	99.5	95.9	96.8	68.0	85.1	87.6	89.2
	Ours	100	84.4	98.5	81.6	88.2	100	97.2	88.4	90.9	88.7	**91.8**

that preset the target scale in advance, such as faster RCNN and MS-VANs mentioned in Table III, densely distributed targets can be covered well.

Results on NWPU VHR-10: We also evaluate the GLNet on NWPU VHR-10 and compare it against the latest methods, encompassing CAD-Net [50], MIF-CNN+ [55], Faster R-CNN [9], F-Faster R-CNN [56], DAPNet [56], and SAPNet [57]. Note that DAPNet and SAPNet focus on the proposal scale issue, which provides a baseline for scale adaptive detectors. MIF-CNN+ and F-Faster R-CNN utilize feature maps in multiple scales and they furnish a baseline for multiscale feature employed detectors. As many approaches do, we stochastically choose 533 positive samples from the total positive samples to be the training set and keep the remaining positive samples as the test set. Table IV shows the experimental results examined on NWPU VHR-10. Our GLNet achieves an mAP of 91.8%, followed by CAD-Net, SAPNet, and DAPNet. In comparison with CAD-Net, ours performs slightly better measured by mAP. The proposed detector upgrades the mAP by a percentage of 2.6 and 4.4, compared with the second and third best detectors SAPNet and DAPNet. In particular, on the categories of ground track field and airplane in NWPU VHR-10, our detector achieves an AP of 100%, which is very difficult to accomplish. Furthermore, our detector also attains a significant improvement in other categories of objects, such as the storage tank, baseball diamond, and vehicle. GLNet achieves an AP of 84.4%, 98.5%, and 90.9%, which are 17.2%, 3.7%, and 5.8% higher than those of the second-best HBB model (SAPNet), respectively. This is great growth in object detection for RSIs. In a word, the proposed method presents a superior detection performance among all these detectors. To view the performance of GLNet, we further reveal several comparative detection results of GLNet against other detectors on NWPU VHR-10, as shown in Fig. 6. The visualization shows that GLNet can commendably detect different objects in challenging situations, reflecting the superiority of GLNet.

Results on DIOR: To explore the generalization ability and robustness of GLNet for different data sets, the recently proposed DIOR RSI data set is examined as well. Table V reports the experimental comparison results on the DIOR data set. Averaged on all categories, an mAP of 70.7% is gained by our model, which ranks number one among all the detectors. The second best detector is RetinaNet, which shows an mAP of 65.7%. Compared with the commonly adopted DOTA data set and NWPU VHR-10 data set, DIOR adds new categories of objects, whereas our model can still exhibit a good detection accuracy on these categories. Specifically, in the category of the dam, an AP of 67.4% is obtained, which leads the second-best detector RetinaNet by a margin of 4.9%. For the categories of overpass and golf course, our model achieves an AP of 62.6% and 81.8%, followed by the CornerNet with an AP of 60.6% and 79.5%, respectively. On the category of the train station, GLNet obtains 65.5% measured by AP, outperformed the second-best detector CornerNet by a large gain of 8.4%, which is a big boosting compared with other baselines. A possible reason is a design of the self-adaptive anchors and the MSP module, which exploits semantic features of the input and builds up a solid foundation for the consequent process. The attention mechanism we employ highlights the target in RSIs and suppresses negative information of complex backgrounds.

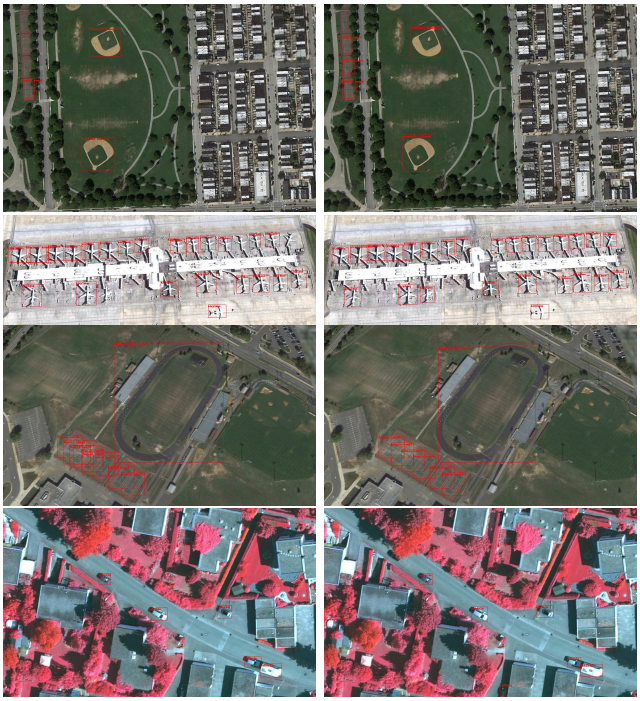


Fig. 6. Visualization of detection results by comparing GLNet against Faster R-CNN on NWPU VHR-10. (a) Faster R-CNN. (b) GLNet.

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TABLE V
QUANTITATIVE COMPARISONS BETWEEN OUR GLNET AND STATE OF THE ARTS ON DIOR. THE PERFORMANCE IS ESTIMATED BY AP(%)

	airplane	airport	stadium	ship	bridge	dam	chimney	harbor	overpass	vehicle	mAP
	ground track field	express way service area	express way toll station	basketball court	baseball field	storage tank	tennis court	train station	golf course	wind mill	
Faster R-CNN [9]	53.6	49.3	73.0	27.7	28.0	62.3	70.9	50.2	50.1	23.6	54.1
	56.9	69.0	55.2	66.2	78.8	39.8	75.2	38.6	68.0	45.4	
Yolo v3 [54]	72.2	29.2	70.6	87.4	31.2	26.9	69.7	44.9	49.7	48.3	57.1
	61.1	48.6	54.4	78.6	74.0	68.7	87.3	29.4	31.1	78.7	
PANet [58]	61.9	70.4	72.9	71.7	38.9	56.6	72.5	41.6	55.8	78.2	63.8
	74.6	68.4	60.0	80.4	71.0	62.3	81.2	54.6	69.0	86.7	
Corner Net [13]	58.8	84.2	70.7	37.6	46.4	64.3	45.3	26.1	60.6	43.0	64.9
	79.5	81.6	76.3	80.8	72.0	45.2	84.0	57.1	79.5	75.9	
Retina Net [59]	53.7	77.3	69.3	71.2	44.1	62.5	72.3	50.7	59.6	45.1	65.7
	74.2	76.2	66.0	81.3	69.0	44.8	81.3	54.2	77.7	83.4	
Ours	62.9	83.2	75.3	72.0	50.5	67.4	79.3	51.8	62.6	43.4	**70.7**
	83.0	86.2	70.9	81.1	72.0	53.7	81.3	65.5	81.8	89.2	

TABLE VI
ABLATION STUDIES ON COMPONENTS OF GLNET ON DOTA, NWPU VHR-10, AND DIOR

Models	Modules			Datasets (mAP)		
	MSP	Clip-LSTM	SA	DOTA	NWPU VHR-10	DIOR
Baseline1	√	√		63.3	90.8	68.6
Baseline2	√		√	63.1	91.2	68.5
Baseline3		√	√	62.2	90.2	68.2
GLNet	√	√	√	65.3	91.8	70.7

C. Ablation Study

1) Effectiveness of Components in GLNet: To examine the contribution of each component to the entire GLNet, we execute ablation studies on three key components MSP, Clip-LSTM, and SA on three data sets. For each data set, four settings are compared, including the proposed approach GLNet, removing the SA module from the proposed method (Baseline1 in Table VI), the proposed method without Clip-LSTM (Baseline2 in Table VI), and the version without MSP module (Baseline3 in Table VI).

Table VI reports the ablation study on three components on DOTA, NWPU VHR-10, and DIOR. For the DOTA data set, compared with the proposed model GLNet, it is easy to find that the performance degrades by removing any component. Specifically, by removing the SA module from the model (Baseline1), the overall detection accuracy is reduced by two percentage points (65.3% versus 63.3%), which demonstrates the effectiveness of SA. This is because SA provides high-quality anchors for the model. When we eliminate the component MSP (Baseline3), the mAP is curtailed from 65.3% to 62.2%, which indicates that the MSP module improves the proposed detection method by about three percentage points. The application of the MSP module can effectively suppress the interference of complex backgrounds. Dismissing Clip-LSTM from GLNet, a decline of 2.2% mAP is observed, which also demonstrates the effectiveness of Clip-LSTM. Similarly, we carry out the ablation study on the NWPU VHR-10 data set, as shown in Table VI. Compared with GLNet, a decrease of 1.0% in mAP can be observed by removing SA, which suggests that SA has a positive influence on the detection of multiscale targets. Baseline3 (without MSP) can also bring a significant drop of 1.6% points, which implies that the attentional mechanism is very useful for object detection in RSIs. Furthermore, a percentage of 0.6 can be observed by cutting off the Clip-LSTM module from the full model. The contribution of the Clip-LSTM examined on NWPU VHR-10 is not as great as that on the DOTA data set, and a possible reason is the limited data amount in the NWPU VHR-10 data set, which makes it difficult to fully exploit the local context relationships. To prove the generalization ability of each component to GLNet, the ablation study is also examined on the DIOR data set, as reflected in Table VI. All the proposed components upgrade the overall detection performance by more than 2% points. Specifically, the SA boosts the overall detection performance by a gain of 2.1% (68.6% versus 70.7%), the MSP module improves the overall

TABLE VII
ABLATION STUDIES ON THE ARCHITECTURE OF CLIP-LSTM

Models	#LSTM units	mAP
Clip-LSTM-2	2	59.3
Clip-LSTM-4	4	60.2
Clip-LSTM-9	9	56.0

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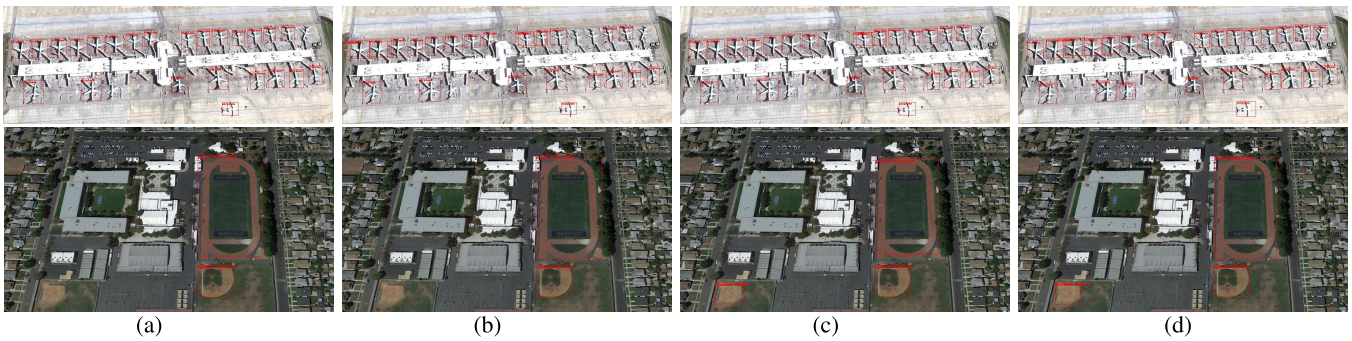
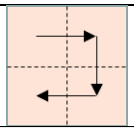
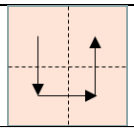
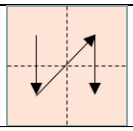
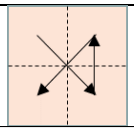
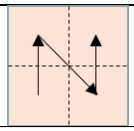
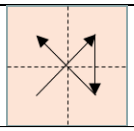
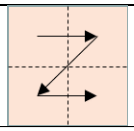
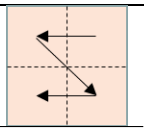


Fig. 7. Visualization of detection results by comparing GLNet with (a) Baseline1, (b) Baseline2, and (c) Baseline3. (d) GLNet.

TABLE VIII
ABLATION STUDIES ON THE ORDER OF LSTM BLOCKS. THE ARROW INDICATES THE CONNECTION FROM ONE LSTM BLOCK TO ANOTHER

Order								
mAP	60.2	59.8	30.3	30.1	29.0	28.7	28.7	27.5

mAP by a percentage of 2.5 (68.2% versus 70.7%), and the Clip-LSTM enhances the overall detection performance by a gain of 2.2% (68.5% versus 70.7%), which is a significant improvement for object detection in RSIs.

Throughout the experimental results of ablation studies on three data sets, SA improves the overall performance on DOTA, NWPU VHR-10, and DIOR by a margin of 2.0%, 1.0%, and 2.1%, respectively. MSP upgrades the performance by an increase of 3.1%, 1.6%, and 2.5%. The Clip-LSTM module shows a gain of 2.2%, 0.6%, and 2.2% on the three data sets. Compared with results on the other data sets, a slightly less growth on the NWPU VHR-10 data set can be observed, a possible reason is the less data it contains, and among these three components, the most closely affected component is Clip-LSTM. Nevertheless, continuous growth in the detection performance can be observed in all three models, which suggests that all three components show strong robustness to different remote sensing data sets.

We also provide a visualization of detection results performed by Baseline1, Baseline2, Baseline3, and GLNet, as presented in Fig. 7. Compared with GLNet, the missing of SA in Baseline 1 shows a relatively weak detection in both large objects (baseball diamond) and small objects (airplanes). Baseline2 shows the worst detection performance for airplanes because of the lacking of Clip-LSTM, and Clip-LSTM specializes in the detection of small targets. MSP focuses on a better representation of targets with various resolutions and the removal of MSP (Baseline3) results in an inferior inspection (inspects fewer airplanes than GLNet), especially for small targets.

2) Analysis of the Clip-LSTM Architecture: To demonstrate the effectiveness of Clip-LSTM, we compare several Clip-LSTM versions with a different number of LSTM units, including two LSTM units (Clip-LSTM-2), four LSTM units (Clip-LSTM-4), and nine LSTM units (Clip-LSTM-9). We randomly select 2300 images from the DOTA data set for training and testing. The experimental results are shown in Table VII. Clip-LSTM with four LSTM units shows the best result. A possible reason is that more LSTM units can express more details of targets. However, in the specific area of remote sensing imaginary, targets are generally very small and a finer partition of the target makes it hard to capture variations of the target itself. Therefore, the performance of Clip-LSTM-9 is not better than that of Clip-LSTM-4. To verify the influence of the order of LSTM blocks in Clip-LSTM, we examine all the permutations of the LSTM blocks (a total of 24 orders for 4 LSTM blocks) on the above data set and reveal the results of the top eight situations in Table VIII. The experimental results show that the clockwise and counterclockwise orders obtain the top two accuracies. Disrupting order impacts semantic representations of objects and deteriorates the recognition performance, which is also consistent with human cognition.

V. CONCLUSION

A global-to-local novel network is developed in this article to detect objects from RSIs, and we concentrate on the issues of the multiscale target, complex backgrounds, and insufficient target information existing in RSIs. The self-adaptive anchor module gathers the semantic features to design anchors instead of manually determining anchors in advance, which assists the detection of multiscale targets in RSIs. The MSP module equips the attention mechanism to learn the global context information, which guides the network to inspect the areas with rich information and eliminates the influence of complex backgrounds. The Clip-LSTM module learns the spatial contexts and local correlations of image features, which provides auxiliary clues for the insufficient target information. Extensive experimental results show that GLNet attains a superior

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performance in all three open data sets against most state-of-the-art methods, which proves the powerful generalization ability and robustness of our model.

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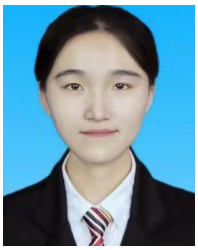
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